DAY 19 – Learning SciKit-Learn Basics

Day 19 introduced one of the most widely used machine learning libraries: **Scikit-Learn**. The session focused on understanding ML workflow, core concepts, and essential tools for building models. Although the main project uses Prophet, learning Scikit-Learn builds strong foundations for future ML projects.

🔹 1. Introduction to Scikit-Learn

We learned that Scikit-Learn provides:

• Simple and efficient ML tools
• Preprocessing utilities
• Regression, classification, clustering algorithms
• Model selection and evaluation techniques
• Easy-to-use API for beginners

It is a standard library for practical ML applications.

🔸 2. Understanding ML Workflow

Scikit-Learn follows a clear workflow:

✓ Load data
✓ Preprocess and clean
✓ Split into train/test sets
✓ Train model
✓ Evaluate performance
✓ Improve using tuning

This structured flow makes ML projects systematic and scalable.

🔸 3. Tools Learned Today

We explored Scikit-Learn features like:

• Train-test split
• StandardScaler for normalization
• LinearRegression model
• Metrics such as R^2 and MAE
• Model persistence using joblib

These tools form the foundation of most ML tasks.

🔸 4. Hands-on Practice

Practical activities included:

• Creating small regression examples
• Fitting simple models
• Predicting new values
• Plotting errors
• Comparing model performance

This helped build confidence with the library.

🔸 5. Importance for Future Projects

Learning Scikit-Learn will help in:

• Building custom ML models
• Working with real-world datasets
• Understanding algorithm behavior
• Experimenting beyond time-series forecasting

It strengthens overall ML understanding.

Key Takeaways from Day 19

• Scikit-Learn simplifies ML tasks
• Workflow clarity improves productivity
• Basic tools are essential for all ML models
• Hands-on practice boosts understanding
• This foundation supports advanced study